Simulating potential exoplanet in quadruple star system HD 98800

Planetary orbits in the region of binary stars

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ABSTRACT

HD98800 is a relatively young quadruple system, about 10 million years old and 45 parsecs away from the Solar System. HD98800 consists of two spectroscopic binary star systems (Aa, Ab and Ba, Bb) orbiting each other with an approximate period of 260 years. In binary system B there exists a gas-rich disk that is orientated in a polar configuration containing well-defined rings, hinting at the possibility of a circumbinary planet's presence. We test the hypothesis of a potential exoplanet's existence via numerical integration methods using Runge-Kutta with varying orbital parameters to find stable configurations of the system. Initial results from simulation data show that the presence of an exoplanet in HD 98800 B is likely and feasible, but requires further investigation.

1. Introduction

HD98800 is a complex quadruple system in the early stages of its life-cycle, with estimates that it is approximately 10 million years old [6] and 45 parsecs away. System A contains a K-type main-sequence star and a second star which does not emit light [7]. System B contains another K-type star and a red dwarf. A debris disk is present in B with orbital inclination of about 90° [8] [4]. The debris disk comprises of an inner ring at 1.5 – 2 AU from the centre of lighter dust particles [3] which may be supported by a planet's resonance. There is a second ring that starts at 5.9 AU and extends up to 13 ± 8 AU comprising of heavier dust particles [4]. The difficulty of modelling such tri-body systems originates from there being no solution describing the interactions [1]. We can only solve such systems numerically, which give rise to errors. A small change in initial conditions can lead to large changes later runtime of the simulation.

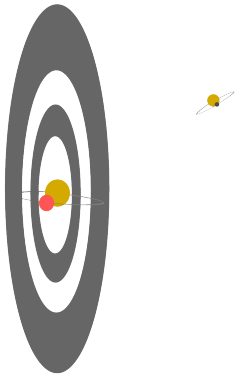


Fig. 1: Rough sketch of the system HD 98800. The binary with the debris disk is HD 98800 B, with disks at inclination 90° to the orbital plane. The second binary is HD 98800 A, located about 40 AU away.

2. Theoretical Method

We use Newton's law of universal gravitational to describe the force felt between two bodies:

$$\boldsymbol{F}_{12} = -\boldsymbol{F}_{21} = -G\frac{M_1 M_2}{|\boldsymbol{r}_{12}|^2}\hat{r}_{12} \tag{1}$$

with $\boldsymbol{F}_{12}$ indicating the force applied to body 1 exerted from body 2, the masses M_1, M_2 of the bodies, the displacement vector $\boldsymbol{r}_{21} = \boldsymbol{r}_2 - \boldsymbol{r}_1$ and gravitational constant G.

In HD 98800 A and B there are 4 discovered celestial objects, and with our proposed circumbinary exoplanet it would make 5. Each body exerts a force on every other body in the system.

The force felt by Body j is the summation of all forces exerted onto it:

$$\sum_{i,\ i\neq j}^{N} \boldsymbol{F}_{ji} = -GM_j \sum_{i,\ i\neq j}^{N} \frac{M_i}{|\boldsymbol{r}_{ji}|^2}\hat{r}_{ji} \tag{2}$$

Using $\boldsymbol{F}_{ji} = M_j\ddot{\boldsymbol{r}}_{ji}$, we divide by M_j to find the acceleration on j:

$$\sum_{i,\ i\neq j}^{N} \ddot{r}_{ji} = -G \sum_{i,\ i\neq j}^{N} \frac{M_i}{|\boldsymbol{r}_{ji}|^2}\hat{r}_{ji} \tag{3}$$

This leads to a second order differential equation, requiring boundary conditions or initial values of the system to solve.

2.1. Initial Conditions

To set the initial conditions of HD 98800 the following approximations are made to simplify calculations: 1. All bodies in the system have zero angular momentum 2. All bodies have equal density over their volume and can be assumed as point masses. 3. A and B, Aa and Ab, Ba and Bb all start at apastron, and the planet starts vertically above the centre of mass in B. By making this approximation, it becomes trivial to compute the initial conditions of all systems. 4. There is only one exoplanet in HD 98800 B, and this planet has no moons and has the same mass

and radius as Jupiter. The planet must have enough mass to clear debris in its orbital path to create gaps in the ring approximately 3AU wide, but must remain small enough to be undetectable. 5. The inclination of the ring is exactly 90° with respect to the plane of orbit in B, and the planes of orbit of A and B are parallel.

The Center of Mass (COM) of HD 98800 A and B, HD 98800 Aa and Ab, HD 98800 Ba and Bb, can be determined using measured data, such as the time period of the orbit, the eccentricity, the semi-major axes and the masses. ([10],). HD98800 A has the following orbital parameters: the mass $M_{Aa} = 0.93R_\odot \pm 0.09$, $M_{Ab} = 0.29R_\odot \pm 0.02$, the semimajor axis $a = 0.86AU \pm 0.02$ and the eccentricity $e = 0.4808 \pm 0.0008$. To find separation from the barycenter of the orbit we need the

Object	Mass ($M_\odot$)	a (AU)	Eccentricity
Aa	0.93 ± 0.09	0.86 ± 0.02	0.4808± 0.0008
Ab	0.29 ± 0.02	0.86 ± 0.02	0.4808± 0.0008
Ba	0.77 ± 0.04	1.01 ± 0.01	0.805± 0.0005
Bb	0.62 ± 0.02	1.01 ± 0.01	0.805± 0.0005

Table 1: Table of Values for the system HD 98800. Semi major axis and Eccentricity are for system A or B and not indivdual objects.

semi-major axis of Aa and Ab relative to the barycenter. It can be computed using product of the total semi-major axis and the mass of the body divided by total mass of the system:

$$a_{Aa} = \frac{aM_{Ab}}{(M_{Aa} + M_{Ab})}, \quad a_{Ab} = \frac{aM_{Aa}}{(M_{Aa} + M_{Ab})} \tag{4}$$

Using the adjusted semi-major axes to find the separation from the barycenter:

$$R_{Aa} = a_{Aa}\frac{(1 - e^2)}{(1 + e\cos(\theta_{Aa}))}, \quad R_{Ab} = a_{Ab}\frac{(1 - e^2)}{(1 + e\cos(\theta_{Ab}))} \tag{5}$$

The eccentricity is the same for both bodies and θ is the true anomaly, describing the position of a body in an elliptical orbit with reference to the closest approach (periapsis). $\theta = 180$ to match the approximations made, placing all bodies at their apastron.

Using the derived positions we then compute the velocities using Isaac Newton's Vis-Viva equation [5] to describe the velocity at a point in an elliptical orbit.

$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right) \tag{6}$$

with M being the total mass ($M_{Aa} + M_{Ab}$), a the semi-major axis (not relative to barycenter), and r is the relative separation between the two bodies. Applying the values computed gives us:

$$v_A^2 = GM\left(\frac{2}{R_{Aa} + R_{Ab}} - \frac{1}{a}\right) \tag{7}$$

The product of the Vis-Viva speed v_A and the mass of the other body divided by total mass of the system gives us the speed of the bodies:

$$v_{Aa} = \frac{v_A M_{Ab}}{(M_{Aa} + M_{Ab})}, \quad v_{Ab} = \frac{v_A M_{Aa}}{(M_{Aa} + M_{Ab})} \tag{8}$$

The values computed, relative to the COM of the system, must be in opposite directions. The objects aligned on the Y axis at apastron positions have velocity $v_{Aa}\hat{x}$ and $-v_{Ab}\hat{x}$ when the orbit is on the XY plane.

This method is the same for HD 98800 Ba and Bb and for HD 98800 A and B, with the added step of finding the semi-major axis of A and B using the time period of their orbit $T_{AB} = 246 \pm 5$ years.

$$a_{AB} = \sqrt[3]{\frac{T_{AB}^2 GM}{4\pi^2}} \tag{9}$$

with M being total mass of HD 98800 $M_A + M_B$, giving a semi-major axis of ≈ 58 AU, which is the expected value.

We can assume the planet orbits around a single body, which would be the COM of HD 98800 B. Doing this and using our earlier approximations for its mass being negligible we find its speed:

$$v_p = \sqrt{\frac{G(M_{Ba} + M_{Bb})}{r}} \tag{10}$$

and if the planet is at 4 AU (between 2 AU and 6 AU) then $v_p = 1.77 \times 10^4 \mathrm{ms}^{-1}$

2.2. Simulation

The simulation is run using the "solve_ivp" function from Python's Scipy Library [9], which is an initial value problem-solving function for ODEs. The integration method used is Runge Kutta, of which we use specifically RK23 in the simulation. Runge Kutta is an explicit and implicit iterative method for finding approximate solutions to nonlinear simultaneous equations [2]. We choose RK23 (order 2,3) to exchange accuracy with simulation speed. It also has the benefit of being less memory intensive, allowing us to run the program for longer time frames. We randomly generate Gaussian distributed values to vary the simulation's initial conditions, with 1σ being equal to the error for the object. This allows the generation of systems that slightly vary from each other, producing different results. By running multiple simulations, we can find systems with good initial conditions (if there are any) and see if there are any trends for success or failure.

Solving numerically requires a step size between each computation. HD 98800 has a very large separation between HD 98800 A and HD 98800 B, having a period of 260 ± 10 years for one orbit. With time step 1, a year of data ($31.5 \times 10^6 \times 30$ data points) takes up approximately 10GiB of memory. For most simulations ran in this project, the step size to year ratio is about 1 second : 2 years. The max step size before the simulation accumulates error rapidly is about 500 seconds, allowing us to visualise 1000 years of orbital data from an old laptop.

A workaround to large step sizes is to save simulation data and use the last time slice as the initial conditions when running the next simulation and repeat. However, this creates a problem of storage space instead of memory space.

The simulation was first tested by simulating the Sun-Earth-Moon interactions for a few years which maintained its orbit for those tests.

3. Results

First we vary the mass of the planet and observe when it leaves its cleared orbit (separation > 6.5 AU) and search for the best mass for this system at 4 AU separation initially. Fig 2

Next we will vary initial separation with mass constant to see when it leaves the system to find the most stable orbital separation for the system. Fig 3,4

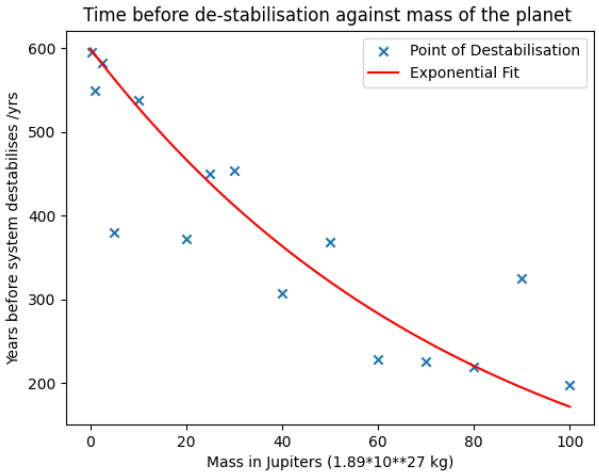


Fig. 2: The figure depicts time taken before the system becomes unstable with a dependence on mass. A smaller mass leads to longer system lifetime. Step size was 400 for each point. Exponential fit is a general fit to show the trend. System is considered unstable when planet's separation to HD98800 Ba or Bb is >6.5AU (in the debris disk)

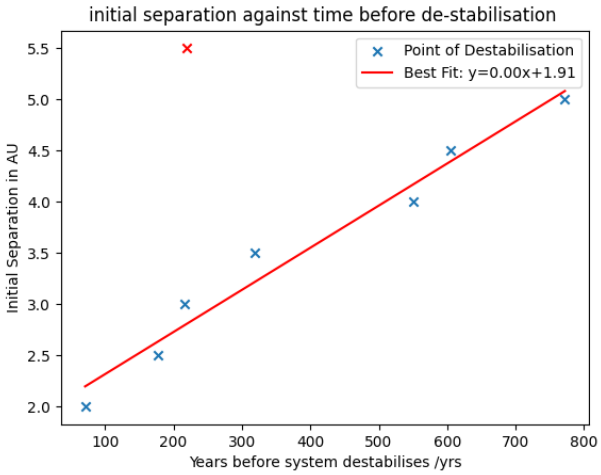


Fig. 4: The figure shows the time when the system becomes unstable (>6.5 AU separation) for different initial separations for the exoplanet of mass equal to Jupiter's. A clear linear trend is seen, with systems remaining stable for longer when they start further from the COM of HD 98800. Red outlier is present because it started close to 6.5 AU, causing it to trigger earlier than other points.

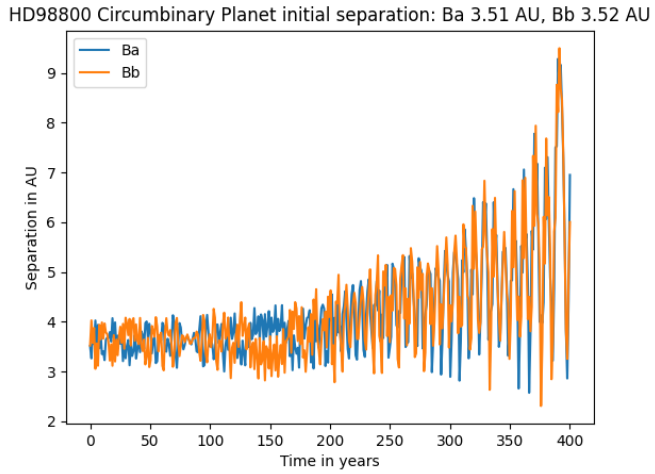


Fig. 3: The figure shows the relative separation between the exoplanet, HD 98800 Ba and HD 98800 Bb over the course of 400 years with timestep 400. It shows overtime the drift in separation as time increases. Exoplanet has initial velocity given by eq 10. Oscillations are caused by planet's orbit around the binary.

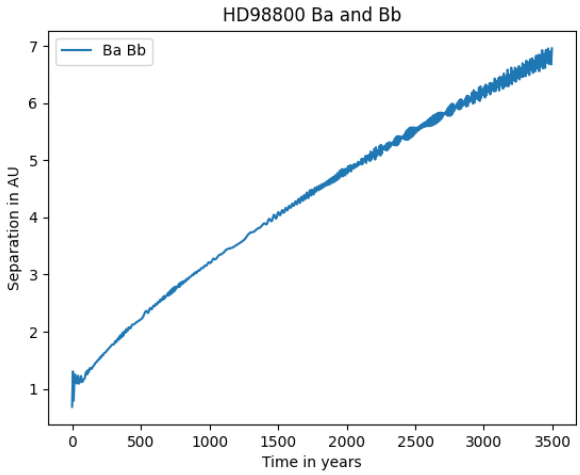


Fig. 5: The figure shows the relative separation between HD 98800 Ba and HD 98800 Bb. Separation slowly increases and orbital frequency also decreases. Timestep is 2000. The exoplanet is not present in this simulation. The mass of each can be found in Table 1, without error bars.

Finally we will visualise the setup by graphing it in 3 Dimensions using colour shade to indicate time of orbit.

4. Discussion of Results

The separation between all objects appear to increase as the simulation progresses. This may be a consequence of a large time step, but it may originate from poor initial conditions. To verify which it may be, future simulations should try varying the true anomaly for a more reliable and natural system. Another possibility may be in our approximations - the assumption of only one planet. If there was a second planet, it may be able to attract the first, keeping its separation in check and stopping it from spiralling out of control.

In Figure 2 it shows the system becomes more unstable for larger planet masses. The mass best suited for an initial separation of 4 AU is $\frac{1}{3}$ the mass of Jupiter. Having a smaller mass means we can assume it to be a satellite and make easy assumptions. But when the mass is no longer negligible it causes the system to become unstable, so our planet has to be light. If the planet was heavy it would be detectable in the orbit of HD 98800 Ba and Bb, so our planet is somewhere the mass of Jupiter still. Each point was computed with a step size of 400, but the value does not change with smaller step sizes.

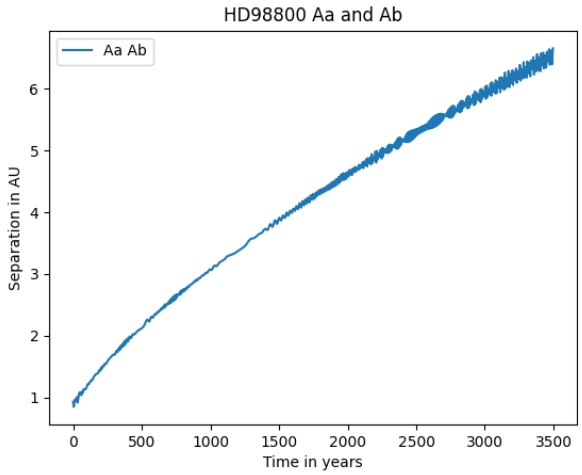


Fig. 6: The figure shows the relative separation between HD 98800 Aa and HD 98800 Ab. Separation slowly increases and orbital frequency also decreases. Timestep is 2000. The exoplanet is not present in this simulation. The mass of each can be found in Table 1, without error bars.

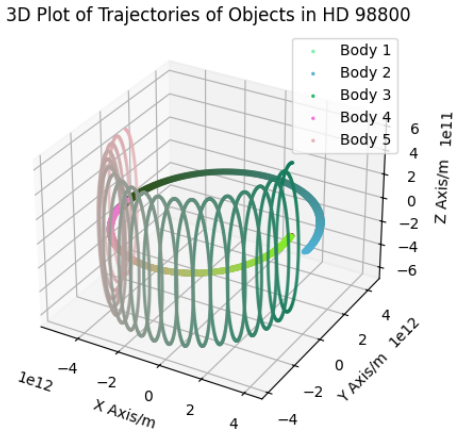


Fig. 7: The plot shows the planet orbiting in a stable configuration. The colour gradient shows the evolution of time, with the colour label representing the initial position. The simulation was ran for 130 years, half an orbital period of HD98800 A and B. Body 1,2 are objects Aa Ab. Body 3,4 are objects Ba Bb, Body 5 is the planet. The planet starts as light pink and as time progresses turns into teal. HD 98800 A starts as teal and becomes dark green. HD98800 B starts as hot pink and becomes lime. The stars are too close to tell them apart. Simulation has values from Table 1 and planet has Mass of Jupiter and initial separation of 4 AU.

In Figure 3 it shows the separation of the planet oscillating and slowly increasing until spiking at 9.5 AU. The planets all drift away from HD 98800 B, leaving their cleared orbit between 2 – 6 AU and then slingshotting out of the system a few hundred years later. A potential fix to this problem would be to introduce a secondary planet, keeping this drift in check whilst still orbiting HD 98800 B. This problem may originate from the drifting of HD 98800 Aa, Ab and HD 98800 Ba and Bb, which slowly separate from each other during simulation runtime.

This apparent drift may be caused by initial conditions being poor or set up incorrectly, or shows that something is missing from these systems. It could be due to our approximations for the simulation. For example, the tidal forces from the stars orbiting so close to each other may cause them to not be perfectly spherical and not spread their mass equally, or there is significant angular momentum that may be affecting its mass or orbit time. It could also be that the mass readings are not perfect, as they have error and for these plots Fig 6, 5 we took them to be the values in Table 1 with no error. The orbit of these stars are incredibly specific and if any parameters are not defined correctly it may cause issues as this. In future simulations, masses Aa Ab and Ba Bb should be varied until a stable system is encountered.

This drift is minimal in the early lifetime of the simulation, resulting in stable orbital configurations for the system. This is not an issue with the simulation code as it was tested with known orbital parameters such as the Sun, Moon and Earth and no anomalies were present.

5. Conclusion

The presence of an exoplanet in HD 98800 B is highly likely and feasible. Its also highly likely multiple exoplanets exist in this system to keep the orbit of other exoplanets more stable, but this needs more investigation.

Simulations show that the planet remains in a stable orbit for >500 years, even with very large time steps. The simulations that run longer than this do not produce good results because of the unprecedented slow increase of separation between the stars in their binary systems HD 98800 A & B. This may be caused by our assumptions or the initial conditions / values are incorrect. With less approximations and a higher accuracy for the orbital parameters in HD 98800 A & HD 98800 B, this drift can be removed when the orbits become stable. Once this limiter has been removed the simulation can run much longer and produce conclusive evidence for an exoplanet or multiple exoplanets.

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